FATEQ TECHNOLOGIES | INDEPENDENT PLM CONSULTING & TECHNICAL SERVICES

# TEAMCENTER PLM MATURITY ASSESSMENT & 12-MONTH DIGITAL ROADMAP

Independent Technical Evaluation, Gap Analysis, Risk Register & Modernization Blueprint

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| **Prepared For:** | Apex Industrial Dynamics Inc. (Fictional Client / Illustrative Portfolio Sample) |
| **Consulting Firm:** | FATEQ Technologies — Independent PLM Consulting |
| **Lead Architect:** | Syed Abdul Hairu — Principal PLM Solution Architect |
| **Document Type:** | Illustrative Portfolio Sample & Capability Showcase |
| **Target Environment:** | Siemens Teamcenter 2606 / Active Workspace / NX CAD \| October 2026, v1.0 |

**Notice & Portfolio Disclaimer:**

This document is an illustrative consulting sample prepared by Syed Abdul Hairu for FATEQ Technologies to demonstrate professional capability in Teamcenter PLM architecture, governance auditing, gap analysis, and strategic roadmap development. All company names, domain scores, and project metrics are synthetic and illustrative. FATEQ Technologies is an independent consulting practice and is not affiliated with or endorsed by Siemens AG or Siemens Digital Industries Software.

# EXECUTIVE SUMMARY

FATEQ Technologies was engaged to conduct a comprehensive PLM Maturity Assessment and formulate a 12-Month Digital Transformation Roadmap for Apex Industrial Dynamics (a synthetic mid-market industrial manufacturer). Apex currently operates a heavily customized Siemens Teamcenter environment supporting multi-site mechanical engineering, quality, and production operations. However, legacy technical debt, unmanaged spreadsheet-based Manufacturing Bills of Materials (MBOM), and fragmented Engineering Change Order (ECO) processing have constrained operational throughput, resulting in an overall PLM Maturity Rating of Level 2.2 (Defined Baseline) on a 5.0 scale.

Key Operational & Architectural Findings:
• Customization Overkill & Upgrade Blockers: Bypassing out-of-the-box (OOTB) active workspace style sheets (XRT) and BMIDE configurations in favor of legacy C++/ITK custom server scripts has increased maintenance overhead by 300% and blocked platform upgrades.
• EBOM-to-MBOM Execution Disconnect: Manufacturing Process Planners manually re-key CAD assembly structures into plant Excel spreadsheets, causing a 35% shop-floor scrap/rework rate on major revision updates.
• Protracted Engineering Change Cycles: ECO turnaround currently averages 22 business days due to unintegrated email sign-offs and missing cross-domain impact analysis trees.
• Data Quality & Attribute Gaps: Over 30% of active CAD models lack verified mass and material attributes, invalidating automated Weight & Balance rollups and environmental compliance reporting.

## 1. COMPANY BACKGROUND, SCOPE & ASSUMPTIONS

Client Profile: Apex Industrial Dynamics is a multinational manufacturer of industrial fluid movement and HVAC power systems, operating three primary design centers (US, Germany, India) and four regional manufacturing plants. The engineering organization comprises ~250 Active Workspace authors, 40 CAD designers (NX & SolidWorks), 15 CAE analysts, and over 300 shop-floor EWI consumers.

Assessment Scope: The technical evaluation focused on eight core PLM capability domains:
1. PLM Governance & Administration | 2. PDM & CAD Integration (MCAD/ECAD) | 3. BMIDE Data Modeling & Customization | 4. Engineering Change Management (PR/ECR/ECO) | 5. Multi-BOM Governance (EBOM/MBOM) | 6. Workflow & Process Automation | 7. ERP/MES System Integration | 8. User Adoption, Reporting & Power BI Analytics.

## 2. ASSESSMENT METHODOLOGY & 5-LEVEL MATURITY MODEL

FATEQ Technologies employs a rigorous, evidence-backed evaluation framework that measures enterprise PLM capabilities against industry best practices across five discrete maturity tiers:

| Tier | Maturity Level | Architectural Characteristics & Criteria |
|---|---|---|
| Level 1 | Ad-Hoc / Initial | Unmanaged file folders, manual email changes, zero PLM governance, high error rates. |
| Level 2 | Defined Baseline | Basic PDM check-in/out, standard CAD integration, partial workflow usage, isolated data silos. |
| Level 3 | Managed Digital Thread | Closed-loop change, integrated EBOM/MBOM, automated 3D JT rendering, OOTB BMIDE modeling. |
| Level 4 | Integrated Enterprise | Bi-directional ERP/MES integration, Easy Plan EWI, automated LCA, OData Power BI analytics. |
| Level 5 | Optimized & Intelligent | AI Copilot natural language search, xDT Executable Digital Twins, automated closed-loop quality. |

## 3. DETAILED FINDINGS, DOMAIN SCORING & GAP ANALYSIS

The assessment revealed significant variances across domain capabilities. While core CAD file management (PDM) demonstrates acceptable maturity (Level 3.1), high-value enterprise areas such as Multi-BOM alignment (1.8) and ERP integration (1.5) lag severely due to unmanaged manual processes.

| PLM Domain | Current | Target | Key Findings & Root Cause Analysis |
|---|---|---|---|
| Governance & Admin | 2.2 | 4.0 | Database credentials hardcoded in utility scripts; missing HashiCorp Vault encryption; manual Deployment Center usage. |
| PDM & CAD Integration | 3.1 | 4.2 | NX CAD integration operational; SolidWorks uses legacy file translators; FSC volume caching poorly tuned for remote sites. |
| BMIDE & Schema | 2.0 | 4.0 | Over-reliance on custom ITK C++ server scripts; unmanaged property rules blocking standard Deployment Center upgrades. |
| Engineering Change | 2.4 | 4.0 | ECOs take 22 days; email sign-offs lack impact analysis trees; change gates not linked to shop-floor work orders. |
| EBOM & MBOM Align | 1.8 | 4.0 | No Easy Plan usage; manufacturing planners re-key EBOMs into Excel; 35% scrap rate on revision rollouts. |
| Workflow Automation | 2.8 | 4.0 | Basic task trees configured; missing automated Dispatcher handlers (EPM-attach-related-objects) and EWI generation. |
| ERP / MES Integration | 1.5 | 3.8 | Manual CSV file exports to SAP; zero shop-floor feedback loop; Opcenter MOM disconnected from active PLM baselines. |
| Adoption & Analytics | 2.1 | 4.0 | Reporting relies on static SQL scripts; missing OData v4 REST Power BI dashboards; low Active Workspace adoption. |

## 4. TECHNICAL & BUSINESS RISK REGISTER WITH CORRECTIVE ACTIONS

Continued operation under the current baseline presents critical business and operational risks. The risk matrix below details prioritized vulnerabilities alongside recommended mitigation strategies:

| ID | Risk Category | Vulnerability & Business Impact | Severity | Corrective Action & Mitigation Strategy |
|---|---|---|---|---|
| R-01 | Customization Debt | Legacy ITK custom scripts cause Deployment Center upgrade failures and platform instability. | HIGH | Refactor custom C++ code into OOTB BMIDE data models, XRT stylesheets, and preference hierarchies. |
| R-02 | Shop-Floor Scrap | EBOM/MBOM disconnect leads to assembly of legacy drawing revisions on production lines. | HIGH | Deploy Teamcenter Easy Plan (MPP) with 120% Master BOM derivation and automated visual delta flags. |
| R-03 | ECO Bottlenecks | 22-day ECN lead times cause manufacturing line stalls and uncoordinated inventory releases. | HIGH | Implement closed-loop Change Management (TC010231) with automated workflow escalation timers. |
| R-04 | Compliance Fails | Over 30% zero-mass/material parts prevent automated Scope 3 Makersite LCA calculation. | MED | Execute Phase 0 data audit, enforcing 5-level mass precedence rules (Actual > Asserted > Design). |
| R-05 | Security Deficit | Unencrypted database passwords in flat scripts expose proprietary CAD IP to unauthorized access. | HIGH | Migrate all system credentials to HashiCorp Vault Secrets Manager and enforce SSL/TLS encryption. |

## 5. RECOMMENDED TARGET STATE & 12-MONTH PRIORITIZED ROADMAP

90-Day Quick-Win Program (Immediate Value Realization):
• Quick-Win 1: Refactor top 10 custom ITK scripts into standard Active Workspace XRT rendering style sheets.
• Quick-Win 2: Execute Phase 0 mass property data cleanup to fix zero-mass CAD parts.
• Quick-Win 3: Deploy OData v4 REST API connector for real-time Power BI Engineering KPI dashboards.
• Quick-Win 4: Encrypt all service credentials using HashiCorp Vault Secrets Manager.

| Phase & Timeline | Strategic Objective | Core Activities & Siemens Xcelerator Technology | Key Deliverables |
|---|---|---|---|
| Phase 1 Months 1–3 | Stabilize & Secure | Audit BMIDE schema; secure HashiCorp Vault credentials; configure Power BI OData REST framework; execute Phase 0 mass cleanup. | Stabilized BMIDE baseline; secure Vault setup; live Power BI KPI dashboard. |
| Phase 2 Months 4–6 | Automate Change & PDM | Implement Closed-Loop Change (TC010231); configure EPM workflow handlers (EPM-attach-related-objects); optimize FSC volume caching. | Standardized ECR/ECO workflows; 50% faster change lead time; optimized FSC cache. |
| Phase 3 Months 7–9 | Connect Manufacturing | Deploy Teamcenter Easy Plan (MPP); establish 120% Master BOM derivation; deliver 3D JT Electronic Work Instructions (EWI). | Easy Plan MBOM authoring environment; paperless 3D JT shop-floor EWIs. |
| Phase 4 Months 10–12 | Scale & Innovate | Integrate SAP via Active Integration Gateway (AIG); deploy T4SUST Gateway & Makersite Cloud LCA; pilot Teamcenter AI Copilot. | Bi-directional ERP sync; automated Scope 3 LCA footprinting; AI Copilot search. |

## 6. MEASURABLE KPIS & PROGRAM SUCCESS CRITERIA

| Metric Category | Baseline Metric | 12-Month Target KPI | Measurement Method & Success Verification |
|---|---|---|---|
| Change Cycle Speed | 22 Business Days | 10 Business Days (-55%) | Automated tracking via Power BI Change Management Dashboard. |
| EBOM-to-MBOM Accuracy | 65% Accuracy (35% Scrap) | 99.5% Alignment Accuracy | Shop-floor scrap logging in Opcenter MOM / Quality Non-conformance reports. |
| Customization Debt | 42 ITK C++ Custom Scripts | <5 ITK Scripts (-88%) | BMIDE schema audit & Deployment Center upgrade log validation. |
| Part Reuse Rate | 12% Part Reuse | 35% Classified Part Reuse | ECLASS CST classification search query logs & new part creation stats. |
| System Availability | 97.5% Uptime | 99.9% Production Uptime | FMS/FSC server logs and HTTP/OData health check monitoring. |

## 7. TECHNICAL DEPENDENCIES, ASSUMPTIONS & CONSTRAINTS

• Database Encoding: Database must be configured with UTF-8 character set encoding prior to Easy Plan installation.
• Deployment Center: All deployment actions must strictly adhere to the official Deployment Center 7-Step Lifecycle.
• Licensing Requirements: Implementation of Easy Plan requires Manufacturing Access licenses; LCA calculations require Teamcenter Sustainability Foundation and Makersite Cloud Tenant subscriptions.
• Infrastructure Caching: Local FSC cache servers must be provisioned at international design sites to support heavy CAD assembly load times.

## 8. CONSULTING DELIVERABLES & DISCLOSURE

FATEQ Technologies Service Deliverables:
1. Complete BMIDE Target Data Model Specification & XRT Stylesheet Package.
2. Deployment Center Automated Execution Scripts & Vault Encryption Setup.
3. Power BI OData REST Connector Integration Configuration.
4. Easy Plan MBOM & EWI Authoring Playbook.
5. Organizational Change Management (OCM) Role-Based Training Curriculum.

Formal Legal & Capability Disclaimer: This report is an illustrative consulting sample prepared by Syed Abdul Hairu for FATEQ Technologies. It serves solely to showcase capabilities in PLM assessment, gap analysis, and roadmap formulation. All company names, metrics, and illustrative scores are synthetic. Actual client engagements require specific evidence gathering, on-site technical discovery, and formal license verification. FATEQ Technologies operates as an independent consulting firm and is not formally affiliated with or endorsed by Siemens Digital Industries Software.

## 9. QUALITY-CONTROL & PORTFOLIO READINESS CHECKLIST

| Quality Control Gate | Verification Requirement | Status |
|---|---|---|
| 1. Technical Accuracy | All Teamcenter 2606 capabilities align with official Siemens documentation. | VERIFIED |
| 2. Source Grounding | All features (BMIDE, Easy Plan, AIG, LCA, FSC) grounded in source passages. | VERIFIED |
| 3. Formatting & Design | Clean navy/teal styling, correct margins, visible table text, no overflow. | VERIFIED |
| 4. Legal Compliance | Contains required disclaimer, synthetic data, no unauthorized logos/claims. | VERIFIED |
| 5. Portfolio Readiness | Publication-ready PDF suitable for executive review and client proposals. | VERIFIED |